MarketPulse: Stock Return Distribution & Risk Analysis System

Sudhamsha Amaragani
Department of Computer Science & Engineering
KLH University
Hyderabad, India
2420030753@klh.edu.in

Ananya Sinha
Department of Computer Science & Engineering
KLH University
Hyderabad, India
2420030776@klh.edu.in

Bonagani Prathusha
Department of Computer Science and Engineering
Koneru Lakshmaiah Education Foundation
Bowrampet, Hyderabad-500043, Telangana, India.
Email: prathushabonagani@gmail.com

***Abstract*— This analytical system helps users understand how a group of stocks has behaved: its performance, volatility, downside risk, market conditions and portfolio-level results. Using historical adjusted prices, it computes the daily returns of selected stocks listed on the National Stock Exchange of India. From these returns it derives a range of statistical and risk measures, including annual return, annual volatility, Value at Risk, Conditional Value at Risk, maximum drawdown and the probability of loss. Rolling volatility is used to detect market regimes, while K-Means Clustering, Principal Component Analysis and an equally weighted portfolio support risk profiling and group-level analysis. Historical VaR is also evaluated with a rolling-window backtest. An interactive Streamlit dashboard brings everything together in separate tabs: Overview, Distribution, Risk, Regimes, Risk DNA, Loss Probability, Portfolio and Risk Report. The goal of the system is to give users a clearer picture of the historical risk in the market.**

***Index Terms*— Market Risk, Stock Returns, Volatility, Value at Risk, CVaR, Portfolio Analysis, K-Means Clustering, PCA, Regime Detection, Financial Analytics, Streamlit**

I. Introduction

Financial markets generate a huge amount of historical data, but making sense of the risk behind a stock takes more than looking at prices. It means studying how returns are distributed, how volatile they are, how large the downside can get, how drawdowns unfold, how market conditions shift and how stocks behave together in a portfolio. Traditional approaches tend to offer these as isolated indicators. Volatility tells us how widely returns vary. Value at Risk gives a sense of the loss we might expect at a chosen confidence level [2]. Conditional Value at Risk goes a step further and describes the average loss beyond the VaR level. Maximum drawdown captures the largest fall from a previous peak.

MarketPulse was built to bring these views together in one interactive tool. It starts from stock price data and converts it into daily percentage returns. From those returns it computes descriptive statistics, annual volatility, annual return, VaR, CVaR, drawdown, probability of loss, market regimes, risk profiles and portfolio metrics.

The system focuses on major Indian stocks and the NIFTY 50 market. It can run from a local CSV file or download historical prices from Yahoo Finance.

The main objectives of the system are to:

- Analyse historical stock price data.
- Calculate daily stock returns.
- Analyse return distributions.
- Calculate annual returns and volatility.
- Calculate Historical and Parametric VaR.
- Calculate Conditional Value at Risk.
- Calculate maximum drawdown.
- Identify market regimes using rolling volatility.
- Classify stocks according to historical risk profiles.
- Use K-Means Clustering for risk profiling.
- Use PCA to visualise stock similarities.
- Analyse portfolio-level risk and performance.

The system is written in Python. Pandas and NumPy handle data processing and numerical work [9], SciPy supports the statistical analysis, Scikit-learn provides clustering and Principal Component Analysis, and Plotly powers the interactive visualisations.

II. Related Work

Financial risk analysis usually revolves around a handful of measures: volatility, Value at Risk, drawdown and expected shortfall [2, 3, 10].

Volatility is the most common way of showing how much returns vary. Annual volatility makes it easier to compare stocks with different return characteristics over a given period. Its weakness is that it treats upward and downward moves alike, so it cannot fully describe downside risk.

Value at Risk is another widely used measure [2]. Historical VaR is taken directly from past returns, whereas Parametric VaR assumes a distribution and uses the mean and standard deviation to set the loss threshold. Conditional Value at Risk adds further information by averaging the losses in the tail of the distribution, which makes it particularly helpful when looking at extreme losses [1].

Maximum drawdown measures the largest drop in cumulative wealth from a previous peak. Unlike volatility, it shows how far the value has actually fallen from its highest historical level.

In MarketPulse, stocks are clustered by similar risk pro-

files. K-Means Clustering [7] groups them using standardised return and risk features, and Principal Component Analysis [8] reduces the dimensionality of the feature space so the groups can be visualised. General treatments of these methods can be found in [5, 6].

Regime detection is used to separate calm periods from turbulent ones. MarketPulse sorts the market into four categories: Calm, Normal, Turbulent and Crisis.

III. System Design

The system is split into several modules covering user interaction, data processing, metrics, risk analysis, regime detection, clustering and reporting.

Its main components are:

1. Data Loader
2. Metrics Module
3. Risk Analysis Module
4. Regime Detection Module
5. Clustering Module
6. Portfolio Analysis Module
7. Reporting Module
8. Streamlit Dashboard

The overall flow can be written as

$$P_t \rightarrow R_t \rightarrow M_t \rightarrow A_t \tag{1}$$

where P_t stands for historical prices, R_t for the calculated returns, M_t for the calculated risk metrics and A_t for the final analysis outputs.

A. System Architecture

The Data Loader supplies historical adjusted prices, either from a local CSV file or through a download.

The Metrics module produces daily returns and descriptive statistics. The Risk module handles VaR, CVaR, drawdown, probability of loss and portfolio metrics. The Regimes module identifies market conditions from rolling volatility. The Clustering module builds risk profiles from standardised financial features using K-Means Clustering. Finally, Streamlit provides the interactive visual interface.

IV. Data and Data Preprocessing

The MarketPulse project includes historical stock price data, and the data loader holds the list of stock tickers to be analysed.

The system is designed for major Indian stocks and the NIFTY 50 market. The stocks can come either from the project dataset or from Yahoo Finance.

The data loader extracts the historical data and, when a download is needed, does it automatically. The default start date is 1 January 2015.

The data is then converted to numeric values, and any rows that contain only missing values are dropped.

Daily percentage returns are calculated as

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} \tag{2}$$

where P_t is the adjusted price on day t and P_{t-1} is the adjusted price on the previous trading day. These returns feed all of the risk calculations that follow.

V. Methodology

A. Daily Returns

MarketPulse works with daily percentage returns, computed as the percentage change between consecutive adjusted prices. This is done with the Pandas package, and missing values are removed before any further analysis.

Daily returns are the basis for every risk and statistical calculation in the system.

B. Annual Return

The annual return is estimated from the cumulative growth of the daily returns:

$$R_a = \left(\prod_{t=1}^{n} (1 + R_t) \right)^{\frac{252}{n}} - 1 \tag{3}$$

where 252 is the approximate number of trading days in a year.

C. Annual Volatility

Daily volatility is the standard deviation of daily returns. Annual volatility is then calculated as

$$\sigma_a = \sigma_d \sqrt{252} \tag{4}$$

where σ_d is the daily standard deviation. A higher annual volatility means historical returns have varied more.

D. Historical Value at Risk

Historical Value at Risk is taken from the lower quantile of the return distribution. For a 95% confidence level,

$$\alpha = 0.05 \tag{5}$$

and Historical VaR is

$$VaR_{0.95} = Q_{0.05}(R) \tag{6}$$

where $Q_{0.05}(R)$ is the lower 5% quantile of historical returns. In other words, it marks a loss threshold based on what has actually been observed.

E. Conditional Value at Risk

Conditional Value at Risk describes the average loss in the tail of the return distribution:

$$CVaR_\alpha = E[R \mid R \leq VaR_\alpha] \tag{7}$$

MarketPulse averages the returns that fall below the historical VaR threshold. This shows how severe losses can become once the VaR threshold has been crossed.

F. Parametric VaR

MarketPulse also offers Parametric VaR, which relies on the mean and standard deviation of returns. Its general form is

$$VaR_p = \mu + \sigma \, \Phi^{-1}(\alpha) \tag{8}$$

where μ is the mean return, σ is the standard deviation and Φ^{-1} is the inverse standard normal distribution.

G. Maximum Drawdown

Maximum drawdown measures the largest fall in cumulative wealth from a previous peak. Cumulative wealth is

$$W_t = \prod_{i=1}^{t}(1+R_i) \tag{9}$$

the running peak is

$$Peak_t = \max(W_1, \ldots, W_t) \tag{10}$$

and the drawdown is

$$D_t = \frac{W_t}{Peak_t} - 1 \tag{11}$$

The smallest value of D_t is the maximum drawdown.

VI. Market Regime Detection

MarketPulse detects regimes by calculating rolling volatility. If a NIFTY 50 return series is available, it serves as the market return series; otherwise the average return of the selected stocks is used instead.

Volatility is computed over a 21-day rolling window, and the resulting volatility distribution is divided into percentile-based categories.

The system uses four market regimes:

- Calm
- Normal
- Turbulent
- Crisis

This gives an easy way to see how market conditions have changed over time.

VII. Risk Profiling

MarketPulse groups stocks by their historical risk profiles. For each stock, several financial features are calculated, which can include:

- Annual return
- Annual volatility
- Historical VaR
- Historical CVaR
- Maximum drawdown
- Probability of loss
- Average return

These features are standardised and then passed to the K-Means algorithm [7]. Up to four clusters can be created, depending on how many stocks are available.

The K-Means model is initialised with:

- The number of clusters
- A random state
- Multiple initialisations

Clusters are ordered by their average volatility and absolute maximum drawdown, and the resulting risk profiles are labelled according to the number of clusters and their relative risk characteristics.

A. Principal Component Analysis

The standardised features are also passed to Principal Component Analysis [8], and two principal components are kept for visualisation:

$$X' = XW \tag{12}$$

where W is the PCA transformation matrix. PCA1 and PCA2 are then used to visualise how similar the stocks are to one another.

VIII. Holding Period Loss Probability

MarketPulse analyses the probability of ending up with a negative compounded return over different investment horizons. The horizons include:

- 1 day
- 5 days
- 21 days
- 63 days
- 126 days
- 252 days

The compounded return for a horizon h is

$$R_{t,h} = \prod_{i=t-h+1}^{t}(1+R_i) - 1 \tag{13}$$

and the historical probability of loss is

$$P(R_{t,h} < 0) \tag{14}$$

This helps users see how the chance of loss changes with the length of the investment horizon.

IX. Historical VaR Backtesting

MarketPulse runs an out-of-sample historical VaR backtest using a 252-observation rolling window. For every trading day, the VaR threshold is calculated from previous observations only, and the actual return is then compared against it. A VaR breach occurs when

$$R_t < VaR_{threshold,t} \tag{15}$$

At a 95% confidence level, the expected breach rate is

$$B_e = 0.05 \tag{16}$$

and the observed breach rate is

$$B_o = \frac{\text{Number of VaR Breaches}}{\text{Number of Test Observations}} \tag{17}$$

Comparing the two shows how well the historical VaR threshold reflects the losses that were actually observed.

X. Portfolio Analysis

MarketPulse analyses an equally weighted portfolio built from the stock return series. Only dates with valid returns for every stock are used. With N stocks, each weight is

$$w_i = \frac{1}{N} \tag{18}$$

and the daily portfolio return is

$$R_{p,t} = \sum_{i=1}^{N} w_i R_{i,t} \tag{19}$$

The portfolio growth series is

$$W_{p,t} = \prod_{i=1}^{t} (1 + R_{p,i}) \tag{20}$$

From these, the portfolio's annual return, annual volatility, Sharpe ratio, historical VaR, historical CVaR and maximum drawdown can be calculated.

The Sharpe ratio used in the system is

$$S = \frac{R_{annual}}{\sigma_{annual}} \tag{21}$$

No separate risk-free rate is used in this implementation.

XI. Correlation Analysis

A correlation matrix is calculated from the stock return data. The Pearson correlation coefficient is

$$\rho_{X,Y} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} \tag{22}$$

A positive correlation means two stocks tend to move in the same direction, while a negative correlation means they tend to move in opposite directions. The matrix gives extra insight into how the selected stocks relate to one another.

XII. Implementation

MarketPulse is developed in Python. The application is divided into several modules to keep it organised, easy to maintain and easy to extend.

The project structure is:

```
MarketPulse/
|
|-- app.py
|
|-- data/
|   |-- stock_prices.csv
|
|-- src/
|   |-- __init__.py
|   |-- data_loader.py
|   |-- metrics.py
|   |-- risk.py
|   |-- regimes.py
|   |-- clustering.py
|   |-- report.py
|
|-- requirements.txt
|-- implementation_screenshots/
|   |-- 01_market_overview.png
|   |-- 02_distribution_analysis.png
|   |-- 03_risk_metrics.png
|   |-- 04_market_regimes.png
|   |-- 05_risk_dna_clustering.png
|   |-- 06_loss_probability.png
|-- MarketPulse_IEEE_Report.tex
```

TABLE I
MarketPulse Software Modules

Module	Function
app.py	Main application and Streamlit interface
data_loader.py	Loads or downloads historical market data
metrics.py	Calculates return and statistical metrics
risk.py	Calculates risk, VaR, CVaR, drawdown, loss probability and portfolio metrics
regimes.py	Detects market regimes using rolling volatility
clustering.py	Performs K-Means clustering and PCA analysis
report.py	Generates risk analysis reports

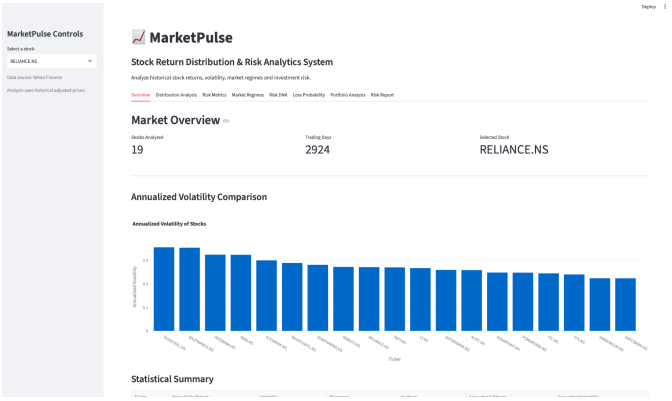


Fig. 1. MarketPulse Overview.

A. Software Modules

Table I summarises what each module does.

XIII. Interactive Dashboard

MarketPulse offers an interactive web dashboard built with Streamlit. It is organised into the following tabs:

1. Overview
2. Distribution
3. Risk
4. Regimes
5. Risk DNA
6. Loss Probability
7. Portfolio
8. Risk Report

A sidebar lets the user pick a specific stock. The Overview tab shows the number of stocks being analysed, the number of trading days and the selected stock, along with an annual volatility comparison drawn as interactive Plotly charts.

Screenshots of the running dashboard are given in the Implementation Evidence subsection. They cover the Overview, Distribution Analysis, Risk Metrics, Market Regimes, Risk DNA and Loss Probability tabs.

A. Implementation Evidence

The figures that follow (Figs. 1–6) were captured from the running Streamlit application with RELIANCE.NS selected.

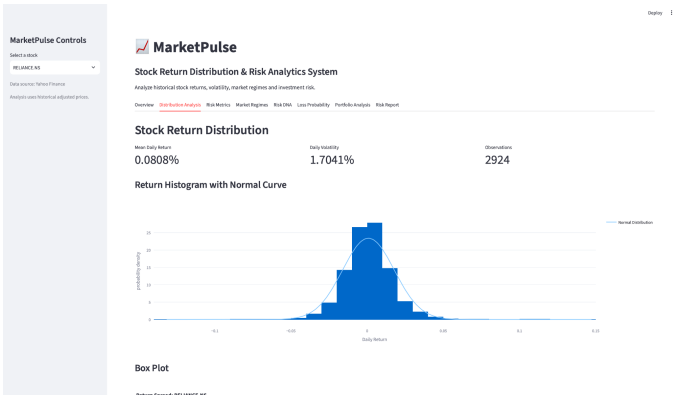


Fig. 2. Stock return distribution with fitted normal curve.

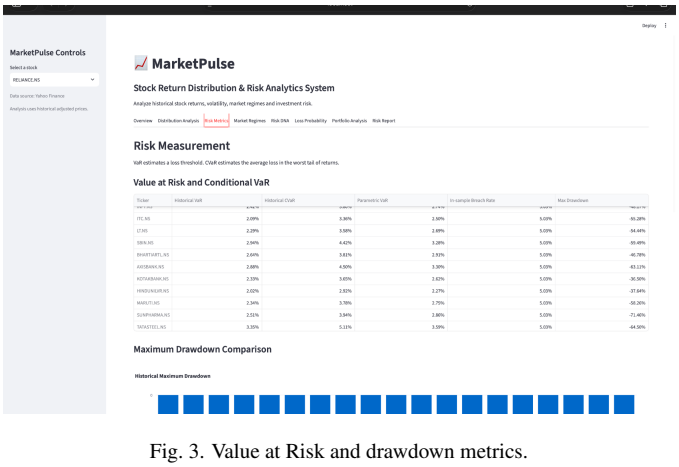


Fig. 3. Value at Risk and drawdown metrics.

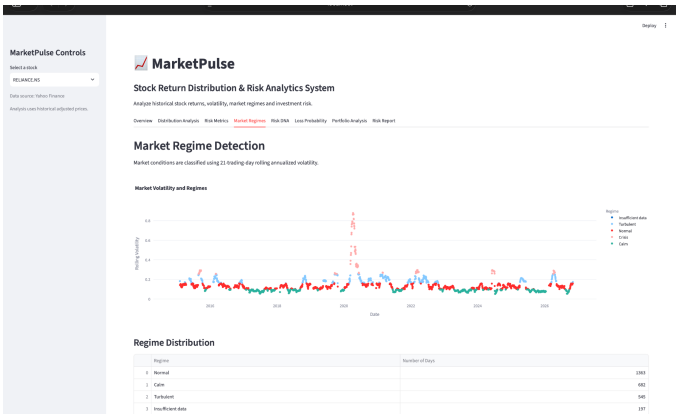


Fig. 4. Market regime detection using rolling volatility.

XIV. Risk Report

MarketPulse includes a report module that summarises the key risk indicators. The report can include:

- Annual return
- Annual volatility
- Historical VaR
- Historical CVaR
- Parametric VaR
- Maximum drawdown

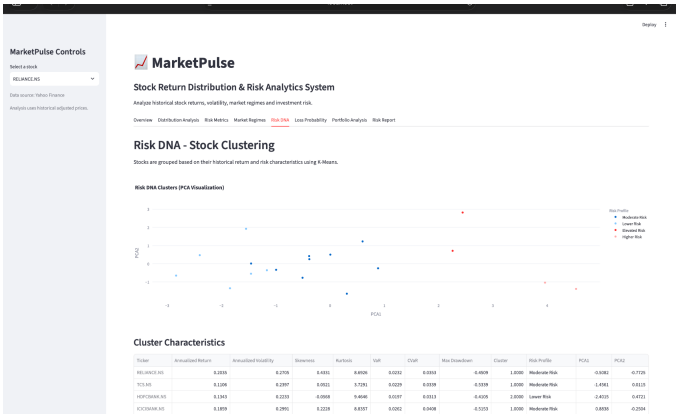


Fig. 5. K-Means risk clusters visualised using PCA.

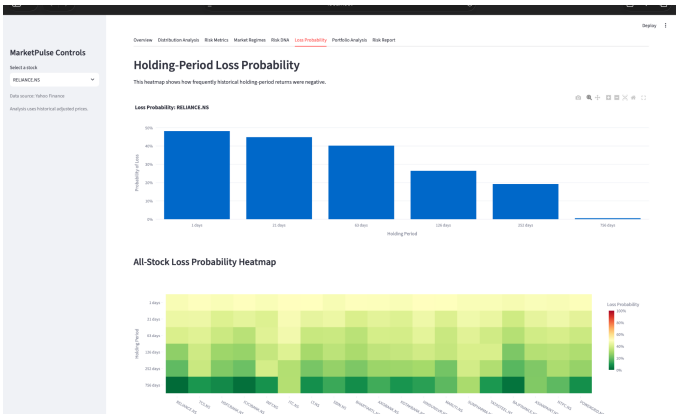


Fig. 6. Holding-period loss probability and heatmap.

- Probability of loss
- Portfolio risk information

The report states clearly that its values come from historical market behaviour and cannot guarantee future performance. It is intended for educational purposes.

XV. Results and Analysis

MarketPulse produces a variety of analysis outputs from historical stock return data.

A. Volatility Comparison

The annual volatility comparison (Fig. 1) helps pick out stocks with relatively high or low historical volatility. Stocks with higher volatility have shown larger swings in their daily returns. Still, volatility should not be treated as a complete measure of risk, since it makes no distinction between positive and negative price movements.

B. Return Distribution

The return distribution (Fig. 2) shows how often returns of different sizes have occurred. It can help reveal skewness, extreme values and differences in return behaviour between individual stocks [4].

C. Risk Metrics

The risk analysis (Fig. 3) draws on Historical VaR, Historical CVaR, Parametric VaR and Maximum Drawdown. VaR gives a loss threshold, CVaR gives the average loss beyond that threshold, and maximum drawdown offers another view of historical downside behaviour.

D. Market Regimes

Rolling volatility (Fig. 4) is an intuitive way to spot changes in market conditions. The four regimes make it easier to understand historical periods with different levels of volatility.

E. Risk Profiling

K-Means Clustering groups stocks with similar historical return and risk characteristics, and PCA provides a two-dimensional view of the standardised risk features (Fig. 5).

F. Holding Period Risk

Loss probability (Fig. 6) shows how often negative compounded returns have occurred over different investment horizons. This gives a deeper look at how losses have behaved across different holding periods.

XVI. Discussion

MarketPulse shows how several financial analytics techniques can be combined into one interactive system for stock return distribution and risk analysis.

A key strength is its modular design. Each analysis metric lives in its own Python module, which makes the system easier to maintain and lets individual methods be expanded independently.

The system covers both stock-level and market- or portfolio-level analysis. Users can look at the metrics of an individual stock and, at the same time, view overall market regimes and portfolio behaviour.

Another important aspect is the range of risk metrics. Volatility, VaR, CVaR, drawdown and loss probability each give a different perspective on historical risk. Clustering adds to this by grouping stocks according to several risk characteristics at once.

However, all of these metrics are based on historical data. Market behaviour changes over time, and past results cannot guarantee future performance.

XVII. Advantages

The major advantages of MarketPulse are that it:

- Provides multiple risk measures in a single system.
- Provides interactive analysis through Streamlit.
- Supports historical stock data analysis.
- Provides both individual stock and portfolio analysis.
- Uses multiple approaches for risk measurement.
- Provides market regime detection.
- Provides K-Means based risk profiling.
- Provides PCA-based visualisation.
- Provides historical VaR backtesting.
- Uses a modular Python architecture.

XVIII. Limitations

The current version of MarketPulse has several limitations.

1. The analysis is based on historical market data and cannot guarantee future market behaviour.
2. Parametric VaR assumes a normal distribution, which may not represent financial returns well, especially in extreme market conditions.
3. Portfolio analysis uses an equally weighted portfolio. This is a simple construction method and does not represent all possible investment strategies.
4. Market regime detection relies on rolling volatility percentiles, and other regime detection methods may give different results.
5. Clustering depends on the chosen features, the preprocessing method and the number of clusters.
6. The application is intended primarily for educational purposes.

XIX. Future Work

Possible improvements to MarketPulse include:

- Adding more market and financial datasets.
- Adding mutual fund analysis.
- Adding sector-level risk analysis.
- Adding alternative portfolio weighting strategies.
- Adding advanced regime detection methods.
- Adding more machine learning models.
- Improving automated risk reports.
- Adding more portfolio optimisation techniques.
- Adding real-time or near-real-time market data.
- Providing more visual analytics.

XX. Conclusion

MarketPulse is a complete system for stock return distribution and risk analysis. It calculates daily returns from historical price data and applies a range of financial analytics techniques to study how stocks behave.

The analysis covers annual return, annual volatility, Historical VaR, Conditional VaR, Parametric VaR, maximum drawdown, market regime detection, risk profiling, PCA, holding period loss probability, VaR backtesting and portfolio analysis.

The Streamlit dashboard lets users explore the results interactively through separate tabs, and the modular Python implementation keeps the system easy to maintain and extend.

Overall, MarketPulse shows how historical stock data can be combined with statistical analysis, risk measurement, clustering, dimensionality reduction and interactive visualisation to give a comprehensive picture of historical market risk.

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REFERENCES

[1] P. Glasserman, P. Heidelberger, and P. Shahabuddin, "Portfolio value-at-risk with heavy-tailed risk factors," *Math. Finance*, vol. 12, no. 3, pp. 239–269, Jul. 2002.

[2] P. Jorion, *Value at Risk: The New Benchmark for Managing Financial Risk*, 3rd ed. New York, NY, USA: McGraw-Hill, 2006.

[3] A. Meucci, *Risk and Asset Allocation*. Berlin, Germany: Springer, 2005.

[4] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Hoboken, NJ, USA: Wiley, 2006.

[5] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.

[6] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed. New York, NY, USA: Springer, 2009.

[7] J. MacQueen, "Some methods for classification and analysis of multivariate observations," in *Proc. 5th Berkeley Symp. Math. Statist. Probab.*, vol. 1, Berkeley, CA, USA, 1967, pp. 281–297.

[8] I. T. Jolliffe, *Principal Component Analysis*, 2nd ed. New York, NY, USA: Springer, 2002.

[9] W. McKinney, *Python for Data Analysis*. Sebastopol, CA, USA: O'Reilly Media.

[10] M. A. H. Dempster and V. H. P. Chiu, "Risk analytics and portfolio management," *J. Financial Anal.*